How the model reads a sign

MediaPipe landmarks and a compact neural network turn static handshapes into A–Z candidates.

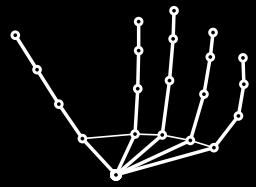
FROM CAMERA FRAME TO LETTER

See the shape. Read the sign.

21 points per hand

Up to two hands

134 input features



LANDMARK SKELETON

MODEL + HAND-COUNT GATE

134 → 256 → 128 → 64 → 26

Landmark MLP; class-weighted training; early stopping.

1 HAND → 1-HAND TRAINING SUPPORT

2 HANDS → 2-HAND TRAINING SUPPORT

A class stays eligible when at least 25% of its training examples match the detected hand count.

17 epochs; best-loss checkpoint at epoch 11.

26 static English letters, including J and K.

THE RECOGNITION LOOP

01 CAMERA	02 MEDIAPIPE	03 134 FEATURES	04 MLP	05 A-Z

HELD-OUT EVALUATION / 3,370 SOURCE IMAGES

76.38%	72.11%	55.07%
ACCURACY WHEN DETECTED	HAND COVERAGE	COVERAGE-ADJUSTED

2,430 / 3,370 test images had detected landmarks. Test set is not signer-separated.

Static alphabet prototype - not connected signing or sentence translation.

If MediaPipe misses a hand, the count can be wrong. Treat output as a suggestion.